HACKIIITD 2026 · ROUND 1 · IDEA SUBMISSION

LegacyBridge

Describe the task in plain English. Get structured data out of systems that never had an API.

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THE PROBLEM

Legacy systems still run the business, and they do not connect to anything new

$370M

average yearly loss per global enterprise from outdated technology

Pega survey of 500+ IT decision-makers

17 hrs

per week the average IT worker spends maintaining legacy systems

ServiceNow and SurveyMonkey Audience survey

92%

of surveyed IT leaders rely on 1 to 20 legacy applications every day

Same Pega survey: 63% use 1–10, 29% use 11–20

WHY THE USUAL FIX FALLS SHORT

Scripts that click through screens become a maintenance job of their own

30–50%

of RPA projects fail outright, according to EY research

EY, as cited by TinyFish Current

25–30%

of RPA total cost is licensing. The rest is implementation, maintenance and support

TinyFish Current

15–20%

of build cost per year is a common maintenance budget for bots on unstable UIs

ThinkInsights playbook (rule of thumb)

THE IDEA

One sentence becomes a recurring data feed

EXAMPLE TASK

“Open the inventory portal, pull Q3 sales by region and post the table to #ops-reports.”

Step 1

Describe

Type the task, pick a destination such as Slack, Sheets, a webhook or SQL, and set a schedule.

Step 2

Run

A cloud browser agent signs in and works through the portal the way an employee would.

Step 3

Deliver

Validated JSON reaches the destination, with a run log showing exactly what happened.

ARCHITECTURE

From prompt to delivered data in five steps

01
Task Builder
Next.js form for the prompt, target and schedule

02
Planner
LangChain turns the prompt into a goal and an output schema

03
TinyFish Agent
Signs in, navigates the portal and extracts the data as JSON

04
Validator
Checks the result against the schema before anything is sent

05
Delivery
Posts to Slack, Sheets, a webhook or a SQL table

UNDER THE HOOD

FastAPI service · job queue and scheduler · Postgres run history · TinyFish Vault for portal logins

WHERE TINYFISH FITS

The browser layer, handled by TinyFish

Goal in, JSON out

The Web Agent API takes a URL and a plain-English goal and returns structured JSON.

No selectors to maintain

The agent reads the live page like a person, so a moved button is less likely to need a code change.

Credentials stay out of prompts

Vault Credentials let the agent sign in without seeing the password. Saved browser profiles keep session and MFA state.

Watch it, scale it

Every run exposes a live streaming URL, and batch endpoints handle many runs at once.

RELIABILITY

The checks an enterprise needs around the agent

Validate

TinyFish's docs note that a COMPLETED run does not mean the goal was met. We check every result against a schema before it leaves.

Approve

Extraction is read-only by default. Any step that writes to a legacy system waits for a person to approve it.

Audit

Each run stores the goal, the live replay link, the extracted data and where it was sent, so any figure can be traced.

Gartner expects over 40% of agentic AI projects to be cancelled by the end of 2027, citing cost, unclear value and weak risk controls. LegacyBridge is designed around the third reason first.

Sources: TinyFish docs; Gartner, via TinyFish Current

WHO IT IS FOR

Teams that live inside old portals

User	Today	With LegacyBridge
Operations heads	Staff copy figures from a portal into spreadsheets every morning	A scheduled run lands the numbers in Sheets or Slack
IT managers and integrators	Months of API work, or a custom script for every legacy tool	One plain-English task per system, no custom connector
B2B SaaS teams	A bespoke connector for each customer’s old system	Customer portals become reachable without new engineering
Logistics, manufacturing and finance	ERP and inventory screens are the only way in	Data from those screens reaches modern tools as JSON

ILLUSTRATIVE SCENARIO

A manufacturer’s morning inventory pull

TODAY

≈ 2 hours

An operations manager reads stock levels from a 15-year-old ERP and emails them, every morning.

WITH LEGACYBRIDGE

Under 1 min

A scheduled run posts the summary to Slack before the shift starts. This is our target run time.

PROTOTYPE SCREENS

Three screens cover the whole flow

1 · Task Builder

Open the inventory portal, pull Q3 sales by region and post the table to #ops-reports.

Slack | Daily 07:30

Run task

2 · Live run

- Signed in with vault login
- Opened Reports, Q3 Sales
- Extracting the table
- Validate, then deliver

3 · Result

Region	Units
North	1,240
South	980
West	1,105

Sample data · sent to #ops-reports

PLAN

What we deliver, and when

Now · Round 1

The idea on paper

Problem, pipeline, reliability layer and three prototype screens.

24-hour finale

A working end-to-end run

Task Builder, planner, TinyFish agent on one demo portal, schema validation, Slack and Sheets delivery, run log.

After the finale

Built for real teams

Approval gates for write actions, per-team isolation, more destinations, run health alerts.

Describe the task. Keep the old system.